EXPERIMENT 6

RESTful API using Node.js and Express

AIM

To develop a RESTful Task Manager API using Node.js and Express to perform CRUD operations.

ALGORITHM

1. Initialize a Node.js project and install Express.
2. Create an Express server and enable JSON parsing.
3. Create routes for POST, GET, PUT and DELETE.
4. Store and modify task data through the routes.
5. Test the endpoints using Postman.

PROGRAM

```
// Install: npm install express
// server.js
const express=require('express');
const app=express();
app.use(express.json());
let tasks=[];

app.post('/tasks',(req,res)=>{
 const task={id:Date.now(),title:req.body.title};
 tasks.push(task); res.status(201).json(task);
});
app.get('/tasks',(req,res)=>res.json(tasks));
app.get('/tasks/:id',(req,res)=>{
 const t=tasks.find(x=>x.id==req.params.id);
 t ? res.json(t) : res.status(404).send('Not found');
});
app.put('/tasks/:id',(req,res)=>{
 const t=tasks.find(x=>x.id==req.params.id);
 if(!t) return res.status(404).send('Not found');
 t.title=req.body.title; res.json(t);
});
app.delete('/tasks/:id',(req,res)=>{
 tasks=tasks.filter(x=>x.id!=req.params.id);
 res.send('Task deleted');
});
app.listen(3000,()=>console.log('Server running on port 3000'));
```

RESULT

Thus, a RESTful API was created successfully using Node.js and Express to perform CRUD operations.

VIVA QUESTIONS

1. What is Node.js?
2. What is Express.js?
3. What is a REST API?
4. What are CRUD operations?
5. What is the purpose of express.json()?